Softplus#

https://pytorch.org/docs/stable/generated/torch.nn.Softplus.html#torch.nn.So

Description:

Applies the Softplus function element-wise. SoftPlus is a smooth approximation to the ReLU function and can be used to constrain the output of a machine to always be positive. For numerical stability the implementation reverts to the linear function when input×β>thresholdinput \times \beta > thresholdinput×β>threshold. beta (float) – the β\betaβ value for the Softplus formulation. Default: 1 threshold (float) – values above this revert to a linear function. Default: 20

Function Signature

```
class torch.nn.Softplus(beta=1.0, threshold=20.0)[source]#
Parameters
Shape:
extra_repr()[source]#
Return type
```

Returns:

Tensor

Code Examples

```
>>> m = nn.Softplus()
>>> input = torch.randn(2)
>>> output = m(input)
```

Detailed Documentation

Documentation

Applies the Softplus function element-wise.

SoftPlus is a smooth approximation to the ReLU function and can be used to constrain the output of a machine to always be positive.

For numerical stability the implementation reverts to the linear function when $input \times \beta > threshold$.

beta (float) – the β value for the Softplus formulation. Default: 1

threshold (float) – values above this revert to a linear function. Default: 20

Input: $(*)$, where $*$ means any number of dimensions.

Output: $(*)$, same shape as the input.

Return the extra representation of the module.